

Trilevel Optimization Model for Competitive Pricing of Electric Vehicle Charging Station Considering Distribution Locational Marginal Price

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Abstract—We study the competitive pricing problem of electric vehicle charging station (EVCS). The EVCS determines its optimal pricing strategy considering the competition from the other EVCS and the impact of charging loads on the distribution locational marginal price (DLMP) of the power network. A trilevel optimization model is proposed, where the upper and middle level capture the profit maximization problems of this EVCS and its competitor respectively, while the lower level represents the optimal power flow problem of distribution network. Due to its computational challenge, this model is reformulated into a single level one, and their equivalence is rigorously proved. Furthermore, we develop an exact algorithm to achieve the global optimal solution, and also prove its finite convergence. Finally, numerical results validate our trilevel model and exact algorithm, and demonstrate their merits in increasing profits for the EVCS.

Index Terms—Trilevel optimization, competitive pricing, electric vehicle, bilevel mixed-integer programming, distribution locational marginal price.

I. INTRODUCTION

ACHIEVING carbon neutrality by the middle of the 21st century has become the common goal of the international

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community. As a consequence, the number of environment-friendly electric vehicles (EV) is soaring in recent years. Over the five year period from 2014 to 2019, the annual average increase of global EV stock is 60% [1].

This trend not only reshapes the transportation sector, but also has great impact on the power grid since huge charging loads are brought along with the rapid penetration of EVs [2]. In particular, for DC fast charging technology that supports large charging power, such as the Tesla supercharger with 250 kW power supply, merely 10 EVs charging simultaneously amounts to 2.5 MW load, which may cause network congestion and affect the distribution locational marginal price (DLMP).

As the coupling point between the transportation network (TN) and power network (PN), the management of EV charging stations (EVCS) is key to maintain the normal operation of these two networks. In fact, the past decade has seen significant research advances towards the optimal management framework for EV charging. However, we focus on a different aspect: the price competition of EVCSs considering their coupling with the PN. For example, if an EVCS lowers its published charging price, more EV users may be attracted from other EVCSs to this station, thus increasing its own charging load while decreasing the loads of others. The resulting load change will in turn affect the DLMP at each EVCS, which is the cost of EVCS for supplying electric energy to EV users. Therefore, when an EVCS determines its optimal pricing decision, both potential competition from other EVCS and the impact on power network's DLMP should be taken into full consideration.

Related Works: The EVCS competition problem has been studied by several researchers during the last few years. In [3], the EVCS price competition problem is formulated as a supermodular game, existence and uniqueness of pure Nash equilibrium is proved. This work is further extended in [4], where the price competition between heterogeneous EVCSs with different capacities is studied using a Stackelberg game. The authors of [5] also consider the competitive pricing problem, but with only two EVCSs on a 1-D system. A Stackelberg game model is proposed and equilibrium conditions are derived analytically. In [6], the optimal siting, sizing

and pricing problem of entrant EVCS considering existing competitors is studied. Here the authors assume that existing players will not take any action in response to the new player. In [7], the optimal placement and spatial price competition problem of EVCSs is studied using the Bayesian game model. In [8], two EV charging service providers are regarded as a duopoly, where both Cournot and Bertrand competition are analyzed. In [9], the authors utilize the deep reinforcement learning approach to learn customer-side information and propose a differentiated pricing mechanism for the multi-service EV charging infrastructure. In [10], the competition between EVCSs is extended to include user preferences and crowdedness, nevertheless, it assumes that charging prices in different EVCSs are identical. In [11], an extensive-form game is introduced to model the planning and pricing competition of EVCSs, and the corresponding subgame-perfect equilibria are found. However, above works all consider EVCSs as price-takers in the power network, i.e., the potential impact of charging loads on DLMP is neglected in each EVCS's optimization problem.

A recently emerging stream of research has explicitly considered the coupling between transportation and power networks through EV charging. For example, in [12], a socially optimal pricing mechanism is proposed to manage the collaborating PN and TN. In [13], a holistic framework is developed to model the coupling transportation-power network (TPN), and a decomposition algorithm is suggested to identify the network equilibrium. This equilibrium is extended to a dynamic one in [14] to capture the temporally-dynamic interactions between PN and TN. In [15], the authors interpret the power-traffic network equilibrium as a potential game, and thereby obtain the market clearing electricity price. In [16], a Stackelberg game based coordinated pricing framework is proposed for the coupled TPN, where the TN operator (TNO) and PN operator (PNO) collaborate as a leader, while EV users are regarded as followers. In [17], the authors proposed a charging pricing strategy based on a bilevel optimization model for the voltage control of distribution networks. In [18], a price-based management method of EV charging services is proposed for smart cities to enhance the operations of coupled TPN and solved by deep reinforcement learning based approach. In [19], the PNO calculates the optimal pricing strategy by jointly minimizing generation cost and the travel cost incurred by EV charging in a coupled TPN. In [20], an integrated energy-traffic system coordinator optimizes both the ride-hailing and charging price to improve the system performance. In [21], the authors proposed a hierarchical optimization problem to account for the optimal decision problem of both EV owners and PN, as well as the optimal pricing problem of charging service operator. This stream of research is generally from the perspective of TNO, PNO or the collaboration of both, which is either non-profit or monopolistic, and is assumed to have access to traffic information such as origin-destination (O-D) pairs of EVs. In contrast, our work is from the perspective of a privately-owned profit-maximizing EVCS that faces oligopolistic competition from its competitor, which may be difficult to obtain private O-D pair information from EV users due to limited authority.

Another category of related literature considers the optimal planning and operation problem from the perspective of EVCS. In [22], the optimal charging pricing problem of public EVCS considering the interdependent operations of TN and PN is formulated as a bilevel optimization problem from the view of charging network operator. In [23], the energy management problem for the coupling operation of EVCSs and distribution system is studied based on supply function game approach. In [24], the authors consider a charging network operator that owns a set of EVCSs, and then design optimal pricing and routing schemes to manage the effect of EV on the power grid. In [25], the stochastic dynamic pricing problem of EVCS is considered to deal with multiple uncertainties such as charging demand volatility, intermittency of renewable energy generation and electricity price fluctuation. In [26], the graph-computing based location planning model of EVCS under both traffic and power grid requirements is proposed to maximize EV charging convenience while ensuring PN's reliability. In [27], the EVCS sizing and placement problem in a coupled TPN is formulated as a mixed integer linear program (MILP), this model takes into account the power line conditioning capability of EVCS, especially as active filters and reactive power compensators. In [28], the authors proposed a coordinated planning method for extreme fast EVCSs considering both TN and PN constraints in order to minimize the investment and electricity costs. In [29], a convexification method is developed to study the EVCS placement problem considering practical aspects such as voltage regulation and protection device upgrade. In [30], a comprehensive planning model is proposed for the expansion of both EVCSs and coupled traffic-electric networks. However, above works only consider one EVCS or one charging service provider, where no price competition between different EVCSs is considered.

Different from the prior arts, we not only consider the price competition of EVCSs, but also recognize the impact of charging loads on the DLMP, thus regarding EVCSs as price-makers in both the transportation and distribution network. Our main contributions are summarized as follows.

- A trilevel model is formulated to simultaneously capture the EVCS price competition and the impact of charging loads on DLMP.
- We propose a novel single-level reformulation, and rigorously prove its equivalence to the original trilevel optimization problem.
- An iterative algorithm is developed to solve the resulting mixed integer single-level program, and its finite convergence to global optimality is also proved.

The remainder of this paper is organized as follows. Section II presents the system model and formulates the trilevel optimization problem. In Section III, we reformulate it into a single level one and propose an exact algorithm to solve it. Numerical results are provided in Section IV, and Section V concludes this paper.

II. SYSTEM MODEL

The system model of the EVCS competitive pricing problem is shown in Fig. 1. Suppose that in a region of interest, e.g., the

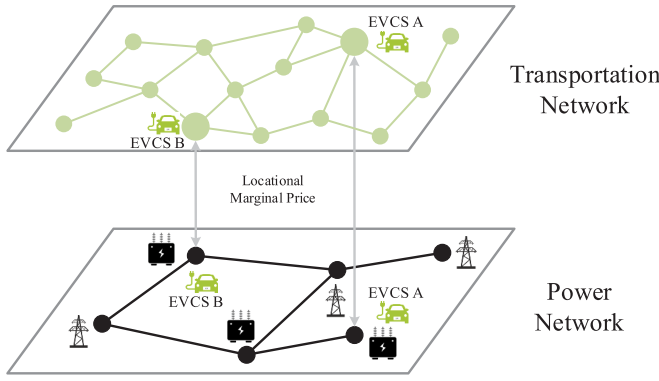


Fig. 1. Coupling transportation-power network.

TABLE I
NOTATIONS

$\mathbb{G} = \{1, 2, \dots, G\}$	The set of G zones
l_a^g, l_b^g	Charging demand from zone g to EVCS A, B
D^g	Intrinsic demand parameter from zone g
d_a^g, d_b^g	Distance from zone g to EVCS A, B
w_d	Load-distance parameter
D_a, D_b	Aggregated demand parameters in (3), (4)
λ_a, λ_b	Charging price of EVCS A, B
α_s^g	Self-price sensitivity parameter of zone g
α_c^g	Cross-price sensitivity parameter of zone g
α_s, α_c	Aggregated price sensitivity parameter
L_a, L_b	Total charging demands of EVCS A, B
μ_a, μ_b	DLMP for EVCS A, B
Φ_a, Φ_b	Profits of EVCS A, B
p_i, q_i	Real and reactive power generation at bus i
d_i^p, d_i^q	Conventional real and reactive power loads
V_i, θ_i	Voltage magnitude and phase angle at bus i
P_{ij}, Q_{ij}	Real and reactive power flow on branch (i, j)

upper layer of Fig. 1, an EVCS A determines its optimal charging price for EVs considering a competitor EVCS B, in order to maximize its profit. The coupling between the transportation network and the lower layer power network is through the DLMP of electric energy provided to the EVCSs. For better understanding of definitions, in Table I, we list all the notations used in this system model section.

A. Model of EVCS

We first divide the region of interest into a set of zones $\mathbb{G} = \{1, 2, \dots, G\}$ according to geographic and residential conditions.¹ We assume that EV charging loads of each zone can be aggregated to the center of this zone. EVCS A locates at zone $a \in \mathbb{G}$, while EVCS B locates at zone $b \in \mathbb{G}$. In each zone, EVs may go to EVCS A or B for charging, and they may also select other alternatives, such as charging at home.

Therefore, based on the linear demand model that is widely used by price competition literature in economics [31]–[33] and the smart grid area [25], charging load demand from zone g to EVCS A l_a^g and demand to EVCS B l_b^g can be represented with the following equations:

$$l_a^g = D^g - w_d d_a^g - \alpha_s^g \lambda_a + \alpha_c^g \lambda_b, \quad (1)$$

$$l_b^g = D^g - w_d d_b^g - \alpha_s^g \lambda_b + \alpha_c^g \lambda_a, \quad (2)$$

¹For example, a zone may represent the neighborhood around a large office building or a resident community.

where $D^g > 0$ is the intrinsic demand parameter of EV charging loads from zone g , d_a^g and d_b^g are the distances of zone g to EVCS A and B respectively, w_d is a positive parameter to model the demand decline as the distances from EVCS become larger. λ_a and λ_b are the charging prices of EVCS A and B, respectively. $\alpha_s^g \geq 0$ is the self-price sensitivity parameter of loads from zone g , which reflects how the change of EVCS charging price can influence its own charging demand. $\alpha_c^g \geq 0$ is the cross-price sensitivity parameter, which reflects how the change of EVCS charging price can influence its competitor's demand. In practice, the parameters in (1) and (2) can be learned from historical data using machine learning based approaches.

Note that a lot of literature related to the management of EVCS in the coupling transportation-power network use flow-based model such as Wardrop equilibrium to derive the charging demand of EVCS. This kind of model generally requires traffic information such as origin-destination pair of EV users [20]. As we have discussed in the introduction, our work is from the perspective of a privately-owned profit-maximizing EVCS, which is different from the literature that is from the perspective of a TNO. Due to limited authority, it may be difficult for an EVCS to obtain full knowledge of traffic flow information like O-D pair. Therefore, we select a zone-based model in (1) and (2) to derive the relationship between EVCS charging demand and charging price.

Intuitively, this demand function means that the charging demand for an EVCS decreases with its own price rise, but increases with the price rise of its competitor. Note that $D^g - w_d d_a^g$ and $D^g - w_d d_b^g$ are both assumed to be positive, such that when the charging service is free, the charging demand will not be zero. By summing up the charging demands from all the zones, the total demands of EVCS A and B become:

$$L_a = \sum_{g \in \mathbb{G}} l_a^g = D_a - \alpha_s \lambda_a + \alpha_c \lambda_b, \quad (3)$$

$$L_b = \sum_{g \in \mathbb{G}} l_b^g = D_b - \alpha_s \lambda_b + \alpha_c \lambda_a, \quad (4)$$

where the constants are defined as $D_a = \sum_{g \in \mathbb{G}} (D^g - w_d d_a^g)$, $D_b = \sum_{g \in \mathbb{G}} (D^g - w_d d_b^g)$, $\alpha_s = \sum_{g \in \mathbb{G}} \alpha_s^g$, $\alpha_c = \sum_{g \in \mathbb{G}} \alpha_c^g$.

As a result, the profits of EVCS A and B can be represented as follows:

$$\Phi_a = (\lambda_a - \mu_a) L_a, \quad (5)$$

$$\Phi_b = (\lambda_b - \mu_b) L_b, \quad (6)$$

where μ_a and μ_b are DLMP at zone a and b respectively, which correspond to the costs for charging EVs. These DLMPs will be discussed with more details in the next subsection.

B. Distribution Network Model

We consider an N -bus distribution network with L branches, where the set of buses is denoted as $\mathcal{N} = \{1, 2, \dots, N\}$, and the set of branches is denoted as $\mathcal{L}$. By abuse of notation, we assume that EVCS A locates at bus $a \in \mathcal{N}$, and EVCS B locates at bus $b \in \mathcal{N}$. A linearized optimal power flow (OPF) model proposed for active distribution networks [34], [35] is selected in this work. Denote the real power generation at bus

$i \in \mathcal{N}$ as p_i , reactive power generation as q_i , conventional real power load that excludes EVCS load as d_i^p , reactive power load as d_i^q , voltage magnitude at bus i as V_i and the voltage phase angle as θ_i . The EVCS load at bus i is defined as L_i , in this case, we have

$$L_i = \begin{cases} L_a, & \text{if } i = a; \\ L_b, & \text{if } i = b; \\ 0, & \text{otherwise.} \end{cases}$$

For a branch $(i, j) \in \mathcal{L}$, where i is the sending bus and j is the receiving bus, its resistance and reactance are assumed to be r_{ij} and x_{ij} respectively. Based on [34, Sec. III], the real and reactive power flow from bus i to bus j can be represented by

$$P_{ij} = \frac{r_{ij}}{r_{ij}^2 + x_{ij}^2} (V_i - V_j) + \frac{x_{ij}}{r_{ij}^2 + x_{ij}^2} (\theta_i - \theta_j), \quad (7)$$

$$Q_{ij} = -\frac{r_{ij}}{r_{ij}^2 + x_{ij}^2} (\theta_i - \theta_j) + \frac{x_{ij}}{r_{ij}^2 + x_{ij}^2} (V_i - V_j). \quad (8)$$

After that we consider the power loss of the distribution network. Notice that the power loss of a branch satisfies the following equations:

$$P_{ji} = -P_{ij} + PL_{ij}, \quad (9)$$

$$Q_{ji} = -Q_{ij} + QL_{ij}, \quad (10)$$

$$PL_{ij} = \frac{P_{ij}^2 + Q_{ij}^2}{V_i^2} r_{ij}, \quad (11)$$

$$QL_{ij} = \frac{P_{ij}^2 + Q_{ij}^2}{V_i^2} x_{ij}, \quad (12)$$

where P_{ji} and Q_{ji} are reverse real and reactive power flows, PL_{ij} and QL_{ij} are real and reactive power losses of this branch. The losses in (11) and (12) can be further linearized via the first-order Taylor approximation, then the reverse power flow equations (9) and (10) become:

$$P_{ji} = -P_{ij} + \frac{\tilde{P}_{ij}^2 + \tilde{Q}_{ij}^2}{\tilde{V}_i^2} r_{ij} + \frac{2r_{ij}\tilde{P}_{ij}}{\tilde{V}_i^2} (P_{ij} - \tilde{P}_{ij}) + \frac{2r_{ij}\tilde{Q}_{ij}}{\tilde{V}_i^2} (Q_{ij} - \tilde{Q}_{ij}) - \frac{2r_{ij}}{\tilde{V}_i^3} (\tilde{P}_{ij}^2 + \tilde{Q}_{ij}^2) (V_i - \tilde{V}_i), \quad (13)$$

$$Q_{ji} = -Q_{ij} + \frac{\tilde{P}_{ij}^2 + \tilde{Q}_{ij}^2}{\tilde{V}_i^2} x_{ij} + \frac{2x_{ij}\tilde{P}_{ij}}{\tilde{V}_i^2} (P_{ij} - \tilde{P}_{ij}) + \frac{2x_{ij}\tilde{Q}_{ij}}{\tilde{V}_i^2} (Q_{ij} - \tilde{Q}_{ij}) - \frac{2x_{ij}}{\tilde{V}_i^3} (\tilde{P}_{ij}^2 + \tilde{Q}_{ij}^2) (V_i - \tilde{V}_i), \quad (14)$$

where $\tilde{P}_{ij}$, $\tilde{Q}_{ij}$, $\tilde{V}_i$ are the linearization points. In practice, various methods can be used to specify these points. In this work, the LinDistFlow model [36] is selected to solve for the initial value of $\tilde{P}_{ij}$, $\tilde{Q}_{ij}$ and $\tilde{V}_i$.

With the previously derived power flow equations, the DLMP at each bus can be solved as the dual variable of an OPF problem, whose constraints are presented as follows:

$$(7), (8), (13), (14) : \rho_{ij}^p, \rho_{ij}^q, \pi_{ij}^p, \pi_{ij}^q \quad \forall (i, j) \\ p_j - d_j^p - L_j = \sum_{k:j \rightarrow k} P_{jk} + \sum_{i:i \rightarrow j} P_{ji} \mu_j \quad \forall j \quad (15)$$

$$q_j - d_j^q = \sum_{k:j \rightarrow k} Q_{jk} + \sum_{i:i \rightarrow j} Q_{ji} \delta_j \quad \forall j \quad (16)$$

$$p_{\min} \leq p_j \leq p_{\max} : \gamma_j^l, \gamma_j^u \quad \forall j \quad (17)$$

$$q_{\min} \leq q_j \leq q_{\max} : \nu_j^l, \nu_j^u \quad \forall j \quad (18)$$

$$V_{\min} \leq V_j \leq V_{\max} : \eta_j^l, \eta_j^u \quad \forall j \quad (19)$$

where (15) and (16) are power balance equations, (17) and (18) model the real and reactive power generation limits at each bus, while (19) represents the voltage limits.

We further denote the set of generators as $\mathcal{N}_g$, and assume that the operation cost of each generator is a convex quadratic function of power generation. Therefore, the cost of generator is defined as

$$C_k(p_k) = a_k p_k^2 + b_k p_k + c_k, \quad \forall k \in \mathcal{N}_g, \quad (20)$$

where a_k , b_k and c_k are non-negative cost coefficients.

The distribution OPF problem can be written as follows:

$$\text{OPF} : \min_{\mathbf{u}} \sum_{k \in \mathcal{N}_g} C_k \\ \text{s.t.} \quad (7)-(8), (13)-(19).$$

Here $\mathbf{u} = (\mathbf{p}, \mathbf{q}, \mathbf{P}, \mathbf{Q}, \mathbf{V}, \boldsymbol{\theta})$ denotes the variables of the OPF problem. By solving this problem, DLMP at each bus i is obtained as the Lagrange multiplier μ_i associated with the real power balance equation (15) [35].

C. Trilevel Optimization Problem

With the EVCS model and distribution network model at hand, in this subsection, we introduce the trilevel optimization formulation for the EVCS competitive pricing problem.

According to the models above, pricing decisions of two EVCSs are coupled not only through the charging demand equations (1)-(2), but also through the OPF constraints. Consequently, when an EVCS determines its optimal pricing strategy, it should take into account the reaction of its competitor as well as their potential impact on the DLMP. Note that we assume the distribution network parameters are available to the EVCS, this can be enabled by existing studies concerning how to extract distribution network information from limited observations [37], [38]. Hence, we extend the leader-follower game structure to a trilevel model depicted in Fig. 2, where the profit maximization problems of EVCS A and B are the upper and middle level respectively, and the OPF problem is the lower level.

The leader-follower pricing structure is a prevalent phenomenon in many industries [39]. In practice, various factors such as market dominance, brand loyalty and technological difference can all lead to this price leadership [40]–[42]. In this paper, we focus on the profit-maximizing problem of the price leader, since the follower's problem is comparably trivial and thus omitted.²

²For a price follower, its profit-maximizing problem can be modeled as a bilevel program. The upper level is that given the leader's price, the follower decides its optimal pricing strategy to maximize its own profit, subject to the lower level where the DLMP is calculated by solving the OPF problem. This bilevel program can be equivalently transformed into a single-level mixed integer nonlinear program by replacing the lower-level OPF problem with its KKT conditions, and then solved by off-the-shelf solvers.

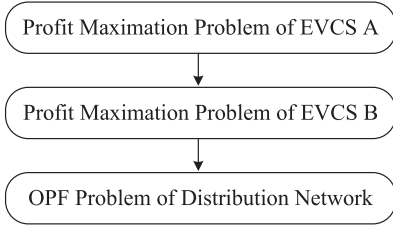


Fig. 2. Trilevel optimization model.

In mathematical terms, the trilevel optimization problem can be formulated as follows:

$$\begin{aligned}
 \mathbf{UL} : & \max_{\lambda_a} \Phi_a, \\
 & \text{s.t. } 0 \leq \lambda_a \leq \lambda_{\max}, \\
 \mathbf{ML} : & \max_{\lambda_b} \Phi_b, \\
 & \text{s.t. } 0 \leq \lambda_b \leq \lambda_{\max}, \\
 \mathbf{LL} : & \min_{\mathbf{u}} \sum_{k \in \mathcal{N}_g} C_k, \\
 & \text{s.t. (7)–(8), (13)–(19),}
 \end{aligned} \quad (21)$$

where $\lambda_{\max}$ is the charging price cap, **UL** and **ML** are profit maximization problems of EVCS A and B respectively, and **LL** is the OPF problem of distribution network.

The solution of Problem (21) constitutes an equilibrium for both EVCS A and B, as well as the PNO. The concept of this competitive pricing equilibrium is introduced as follows:

Definition 1: The decision variables of EVCS A, B and PNO ($\lambda_a^*, \lambda_b^*, \mathbf{u}^*$) constitutes a competitive pricing equilibrium if the following inequalities hold:

$$\begin{aligned}
 \Phi_a(\lambda_a^*, \lambda_b^*, \mathbf{u}^*) & \geq \Phi_a(\lambda_a, \lambda_b^*, \mathbf{u}^*), \quad \forall \lambda_a \in \Lambda_a, \\
 \Phi_b(\lambda_a^*, \lambda_b^*, \mathbf{u}^*) & \geq \Phi_b(\lambda_a^*, \lambda_b, \mathbf{u}^*), \quad \forall \lambda_b \in \Lambda_b, \\
 \sum_{k \in \mathcal{N}_g} C_k(\lambda_a^*, \lambda_b^*, \mathbf{u}^*) & \leq \sum_{k \in \mathcal{N}_g} C_k(\lambda_a^*, \lambda_b^*, \mathbf{u}), \quad \forall \mathbf{u} \in \mathcal{U},
 \end{aligned}$$

where Λ_a , Λ_b and $\mathcal{U}$ are feasible sets of λ_a , λ_b and $\mathbf{u}$.

Remark 1: This definition implies that at equilibrium, none of the three players has incentive to change their behavior unilaterally. The optimal solution ($\lambda_a^*, \lambda_b^*, \mathbf{u}^*$) of Problem (21) is also a competitive pricing equilibrium. The reason is that when (λ_a^*, λ_b^*) are fixed, $\mathbf{u}^*$ is the optimal solution of PNO's problem **LL**. When ($\lambda_a^*, \mathbf{u}^*$) are fixed, λ_b^* maximizes EVCS B's profit Φ_b . When ($\lambda_b^*, \mathbf{u}^*$) are fixed, λ_a^* maximizes the profit Φ_a of EVCS A. Therefore, deviating from the optimal solution of Problem (21) can only bring welfare loss to these three players, which conforms with Definition 1.

Solving a trilevel optimization problem exactly is always a challenging task, thus in the next section, we pursue a tractable algorithm for it.

III. PROBLEM TRANSFORMATION & ALGORITHM DESIGN

In this section, at first the trilevel optimization problem is transformed into an equivalent mixed integer bilevel program, which is further transformed equivalently into a single-level optimization problem based on the value function approach.

Finally an exact algorithm is proposed to achieve the global optimal solution.

A. Equivalent Bilevel Problem

Note that the OPF problem is convex since its objective function is convex and its constraints are linear. In practice, there usually exists an interior point in the feasible region of OPF problem that satisfies the inequalities strictly, in other words, the Slater's condition holds [43]. Therefore, the OPF problem can be equivalently represented by its KKT conditions:

$$\begin{aligned}
 & (7)–(8), (13)–(16), \\
 & 2a_j p_j + b_j - \mu_j + \gamma_j^u - \gamma_j^l = 0, \quad \forall j \in \mathcal{N}
 \end{aligned} \quad (22)$$

$$-\delta_j + v_j^u - v_j^l = 0, \quad \forall j \in \mathcal{N} \quad (23)$$

$$\begin{aligned}
 & \mu_i - \rho_{ij}^p - \pi_{ij}^p \left(1 - \frac{2\tilde{P}_{ij} r_{ij}}{\tilde{V}_i^2} \right) + \pi_{ij}^q \frac{2\tilde{P}_{ij} x_{ij}}{\tilde{V}_i^2} = 0, \\
 & \quad \forall (i, j) \in \mathcal{L}
 \end{aligned} \quad (24)$$

$$\begin{aligned}
 & \delta_i - \rho_{ij}^q + \pi_{ij}^p \frac{2\tilde{Q}_{ij} r_{ij}}{\tilde{V}_i^2} - \pi_{ij}^q \left(1 - \frac{2\tilde{Q}_{ij} x_{ij}}{\tilde{V}_i^2} \right) = 0, \\
 & \quad \forall (i, j) \in \mathcal{L}
 \end{aligned} \quad (25)$$

$$\mu_j - \pi_{ij}^p = 0, \quad \forall (i, j) \in \mathcal{L} \quad (26)$$

$$\delta_j - \pi_{ij}^q = 0, \quad \forall (i, j) \in \mathcal{L} \quad (27)$$

$$\begin{aligned}
 & - \sum_{i:i \rightarrow j} \frac{\rho_{ij}^p r_{ij} + \rho_{ij}^q x_{ij}}{r_{ij}^2 + x_{ij}^2} + \sum_{k:j \rightarrow k} \frac{\rho_{jk}^p r_{jk} + \rho_{jk}^q x_{jk}}{r_{jk}^2 + x_{jk}^2} \\
 & - \sum_{k:j \rightarrow k} \frac{2(\tilde{P}_{jk}^2 + \tilde{Q}_{jk}^2)(\pi_{jk}^p r_{jk} + \pi_{jk}^q x_{jk})}{\tilde{V}_j^3} + \eta_j^u - \eta_j^l = 0, \\
 & \quad \forall j \in \mathcal{N}
 \end{aligned} \quad (28)$$

$$\begin{aligned}
 & - \sum_{i:i \rightarrow j} \frac{\rho_{ij}^p x_{ij} - \rho_{ij}^q r_{ij}}{r_{ij}^2 + x_{ij}^2} + \sum_{k:j \rightarrow k} \frac{\rho_{jk}^p x_{jk} - \rho_{jk}^q r_{jk}}{r_{jk}^2 + x_{jk}^2} = 0, \\
 & \quad \forall j \in \mathcal{N}
 \end{aligned} \quad (29)$$

$$\gamma_j^u \geq 0 \perp p_j \leq p_{\max}, \quad \gamma_j^l \geq 0 \perp p_j \geq p_{\min}, \quad \forall j \in \mathcal{N} \quad (30)$$

$$v_j^u \geq 0 \perp q_j \leq q_{\max}, \quad v_j^l \geq 0 \perp q_j \geq q_{\min}, \quad \forall j \in \mathcal{N} \quad (31)$$

$$\eta_j^u \geq 0 \perp V_j \leq V_{\max}, \quad \eta_j^l \geq 0 \perp V_j \geq V_{\min}, \quad \forall j \in \mathcal{N}. \quad (32)$$

Since adding complementarity conditions (30)–(32) into the constraints of optimization problem **ML** will lead to a mathematical program with equilibrium constraints (MPEC) that is difficult to tackle with [44], we introduce a new set of binary variables based on the Big-M approach, and transform the constraints (30)–(32) into an equivalent set of inequalities:

$$0 \leq -p_j + p_{\max} \leq (1 - \kappa_{p,j}^u) M_p, \quad \forall j \in \mathcal{N} \quad (33)$$

$$0 \leq \gamma_j^u \leq \kappa_{p,j}^u M_p, \quad \forall j \in \mathcal{N} \quad (34)$$

$$0 \leq p_j - p_{\min} \leq (1 - \kappa_{p,j}^l) M_p, \quad \forall j \in \mathcal{N} \quad (35)$$

$$0 \leq \gamma_j^l \leq \kappa_{p,j}^l M_p, \quad \forall j \in \mathcal{N} \quad (36)$$

$$0 \leq -q_j + q_{\max} \leq (1 - \kappa_{q,j}^u) M_p, \quad \forall j \in \mathcal{N} \quad (37)$$

$$0 \leq v_j^u \leq \kappa_{q,j}^u M_p, \quad \forall j \in \mathcal{N} \quad (38)$$

$$0 \leq q_j - q_{\min} \leq (1 - \kappa_{q,j}^l) M_p, \forall j \in \mathcal{N} \quad (39)$$

$$0 \leq v_j^l \leq \kappa_{q,j}^l M_p, \forall j \in \mathcal{N} \quad (40)$$

$$0 \leq -V_j + V_{\max} \leq (1 - \xi_j^u) M_p, \forall j \in \mathcal{N} \quad (41)$$

$$0 \leq \eta_j^u \leq \xi_j^u M_p, \forall j \in \mathcal{N} \quad (42)$$

$$0 \leq V_j - V_{\min} \leq (1 - \xi_j^l) M_p, \forall j \in \mathcal{N} \quad (43)$$

$$0 \leq \eta_j^l \leq \xi_j^l M_p, \forall j \in \mathcal{N} \quad (44)$$

where κ_p^u , κ_p^l , κ_q^u , κ_q^l , ξ^u and ξ^l are binary vectors, M_p is a sufficient large number.

Naturally, we can replace the OPF problem in the lower level with its KKT conditions (7)-(8), (13)-(16), (22)-(29), (33)-(44), and add them to the constraints of middle level problem. Thus, we have transformed the original trilevel problem into an equivalent bilevel problem as follows:

$$\begin{aligned} \text{UL : } \max_{\lambda_a} & (\lambda_a - \mu_a) L_a, \\ \text{s.t. } & 0 \leq \lambda_a \leq \lambda_{\max}, \\ \text{MLL : } \max_{\lambda_b, \mathbf{u}} & (\lambda_b - \mu_b) L_b, \\ \text{s.t. } & 0 \leq \lambda_b \leq \lambda_{\max}, \\ & (7)-(8), (13)-(16), \\ & (22)-(29), (33)-(44). \end{aligned} \quad (45)$$

As there exist binary variables in (33)-(44), the resulting problem (45) is a bilevel mixed integer nonlinear program (BMINLP) with discrete variables in the lower level. The bilevels programs are inherently difficult to solve, even the simplest of its kind, i.e., the bilevel linear program with no discrete variable is theoretically NP-hard [45]. Therefore, inspired by the recently developed value function based approach [46], we reformulate the BMINLP and propose an exact algorithm for it in the following subsections.

B. Equivalent Single-Level Problem

In this subsection, we develop an equivalent single-level reformulation for the BMINLP (45). First, a set of definitions are introduced to facilitate the presentation.

Let $\mathbf{x} = \lambda_a$, $\mathbf{y} = (\lambda_b, \mathbf{u})$, then BMINLP (45) can be rewritten in the following compact form:

$$\text{UL : } \max_{\mathbf{x}, \mathbf{y}} \Phi_a(\mathbf{x}, \mathbf{y}), \quad (46a)$$

$$\text{s.t. } \mathbf{x} \in \Omega_x, \quad (46b)$$

$$\text{MLL : } \max_{\mathbf{y}} \Phi_b(\mathbf{x}, \mathbf{y}), \quad (46c)$$

$$\text{s.t. } \mathbf{y} \in \Omega_y(\mathbf{x}), \quad (46d)$$

where $\Omega_x = \{\mathbf{x} | 0 \leq \lambda_a \leq \lambda_{\max}\}$, and $\Omega_y(\mathbf{x}) = \{\mathbf{y} | 0 \leq \lambda_b \leq \lambda_{\max}, (7)-(8), (13)-(16), (22)-(29), (33)-(44)\}$ is the feasible set of $\mathbf{y}$ when $\mathbf{x}$ is fixed. Let

$$\Omega = \{(\mathbf{x}, \mathbf{y}) | \mathbf{x} \in \Omega_x, \mathbf{y} \in \Omega_y(\mathbf{x})\} \quad (47)$$

i.e., Ω is obtained by combining the constraint (46b) and (46d). However, satisfying Ω alone is not enough for an $(\mathbf{x}, \mathbf{y})$ to be a feasible solution to the BMINLP. This is due to (46c) which

requires $\mathbf{y}$ must be an optimal solution of problem **MLL** given $\mathbf{x}$, i.e.,

$$\mathbf{y} \in \operatorname{argmax} \{\Phi_b | \mathbf{y} \in \Omega_y(\mathbf{x})\}. \quad (48)$$

After that, let

$$\Omega_y = \bigcup_{\mathbf{x} \in \Omega_x} \Omega_y(\mathbf{x}) \quad (49)$$

be the set of all feasible variables of the **MLL** problem. For every element $\hat{\mathbf{y}} \in \Omega_y$, two constants are defined:

$$\hat{\Delta}_a^{\hat{\mathbf{y}}} = D_a + \alpha_c \hat{\lambda}_b - \hat{p}_a + d_a^p + \sum_{k:a \rightarrow k} \hat{P}_{ak} + \sum_{i:i \rightarrow a} \hat{P}_{ai}, \quad (50)$$

$$\hat{\Delta}_b^{\hat{\mathbf{y}}} = -D_b + \alpha_s \hat{\lambda}_b + \hat{p}_b - d_b^p - \sum_{k:b \rightarrow k} \hat{P}_{bk} - \sum_{i:i \rightarrow b} \hat{P}_{bi}, \quad (51)$$

where d_a^p and d_b^p are the conventional real loads at bus a and b . Then we can prove that BMINLP (46) is equivalent to a single-level reformulation, as shown in Theorem 1.

Theorem 1: The BMINLP is equivalent to the following single-level optimization problem:

$$\text{SLP : } \max_{\mathbf{x}, \mathbf{y}, \mathbf{z}} \Phi_a(\mathbf{x}, \mathbf{y}), \quad (52a)$$

$$\text{s.t. } (\mathbf{x}, \mathbf{y}) \in \Omega, \quad (52b)$$

$$\alpha_s \lambda_a > \hat{\Delta}_a^{\hat{\mathbf{y}}} - M_a + M_a \zeta_a^{\hat{\mathbf{y}}+}, \forall \hat{\mathbf{y}} \in \Omega_y, \quad (52c)$$

$$\alpha_s \lambda_a < \hat{\Delta}_a^{\hat{\mathbf{y}}} + M_a - M_a \zeta_a^{\hat{\mathbf{y}}-}, \forall \hat{\mathbf{y}} \in \Omega_y, \quad (52d)$$

$$\alpha_c \lambda_a > \hat{\Delta}_b^{\hat{\mathbf{y}}} - M_b + M_b \zeta_b^{\hat{\mathbf{y}}+}, \forall \hat{\mathbf{y}} \in \Omega_y, \quad (52e)$$

$$\alpha_c \lambda_a < \hat{\Delta}_b^{\hat{\mathbf{y}}} + M_b - M_b \zeta_b^{\hat{\mathbf{y}}-}, \forall \hat{\mathbf{y}} \in \Omega_y, \quad (52f)$$

$$\begin{aligned} \Phi_b(\mathbf{x}, \mathbf{y}) & \geq \Phi_b(\mathbf{x}, \hat{\mathbf{y}}) - M_a (\zeta_a^{\hat{\mathbf{y}}+} + \zeta_a^{\hat{\mathbf{y}}-}) \\ & - M_b (\zeta_b^{\hat{\mathbf{y}}+} + \zeta_b^{\hat{\mathbf{y}}-}), \forall \hat{\mathbf{y}} \in \Omega_y, \end{aligned} \quad (52g)$$

where M_a and M_b are sufficiently large constants, $\zeta_a^{\hat{\mathbf{y}}+}$, $\zeta_a^{\hat{\mathbf{y}}-}$, $\zeta_b^{\hat{\mathbf{y}}+}$ and $\zeta_b^{\hat{\mathbf{y}}-}$ are binary variables, $\mathbf{z} = (\zeta_a^{\hat{\mathbf{y}}+}, \zeta_a^{\hat{\mathbf{y}}-}, \zeta_b^{\hat{\mathbf{y}}+}, \zeta_b^{\hat{\mathbf{y}}-})$ for all $\hat{\mathbf{y}} \in \Omega_y$.

Proof: Please refer to Appendix A for the proof. ■

Despite the fact that the original trilevel problem has been reformulated into a single level one, the resulting problem (52) is still challenging to solve. This is because the set Ω_y can be enormous, leading to an exponential number of constraints.

C. Constraint Generation Algorithm

In order to fully solve the trilevel competitive pricing problem, we propose a constraint generation algorithm in this subsection. Before presenting the algorithm, we introduce an assumption first.

Assumption 1: Prices published by both EVCSs are integer multiples of a certain unit s , that is, $\lambda_a = \bar{\lambda}_a s$ and $\lambda_b = \bar{\lambda}_b s$.

Remark 2: Assumption 1 is a reasonable assumption since in practice, all kinds of prices can be represented as integer multiples of a certain unit. For example, $0.51\$ = 51 \times 0.01\$$, where $0.01\$$ is the price unit.

According to Theorem 1, solving problem **SLP** exactly will achieve the optimal solution for the original trilevel problem. Instead, solving a relaxed version of **SLP** as follows yields an upper bound of the trilevel problem's optimal value.

$$\text{RSLP : } \max_{\mathbf{x}, \mathbf{y}, \mathbf{z}} \Phi_a(\mathbf{x}, \mathbf{y}), \quad (53a)$$

Algorithm 1 Constraint Generation Algorithm

Initialization: Set the counter $k = 1$, upper bound $UB = \infty$, lower bound $LB = -\infty$. Select an initial sample set $\hat{\Omega}_{y,1} \subseteq \Omega_y$.

- 1: **repeat**
- 2: Solve problem **RSLP** (53), obtain its optimal solution $(\mathbf{x}_k, \mathbf{y}_k^r, \mathbf{z}_k)$, and update the upper bound $UB = \Phi_a(\mathbf{x}_k, \mathbf{y}_k^r)$.
- 3: Given $\mathbf{x}_k$, solve problem **MLL** in (46), obtain its optimal solution $\mathbf{y}_k^m$.
- 4: Set $\hat{\Omega}_{y,k+1} = \hat{\Omega}_{y,k} \cup \mathbf{y}_k^m$.
- 5: **if** $\Phi_b(\mathbf{x}_k, \mathbf{y}_k^r) = \Phi_b(\mathbf{x}_k, \mathbf{y}_k^m)$ **then**
- 6: Update $LB = UB = \Phi_a(\mathbf{x}_k, \mathbf{y}_k^r)$, and $(\mathbf{x}_k, \mathbf{y}_k^r)$ is the optimal solution of BMINLP.
- 7: **else if** $\Phi_a(\mathbf{x}_k, \mathbf{y}_k^m) > LB$ **then**
- 8: Update $LB = \Phi_a(\mathbf{x}_k, \mathbf{y}_k^m)$.
- 9: **else**
- 10: LB is unchanged.
- 11: **end if**
- 12: Update $k = k + 1$.
- 13: **until** $UB \leq LB$

$$\text{s.t. } (\mathbf{x}, \mathbf{y}) \in \Omega, \quad (53b)$$

$$\alpha_s \lambda_a > \hat{\Delta}_a^{\hat{y}} - M_a + M_a \zeta_a^{\hat{y}+}, \forall \hat{y} \in \hat{\Omega}_y, \quad (53c)$$

$$\alpha_s \lambda_a < \hat{\Delta}_a^{\hat{y}} + M_a - M_a \zeta_a^{\hat{y}-}, \forall \hat{y} \in \hat{\Omega}_y, \quad (53d)$$

$$\alpha_c \lambda_a > \hat{\Delta}_b^{\hat{y}} - M_b + M_b \zeta_b^{\hat{y}+}, \forall \hat{y} \in \hat{\Omega}_y, \quad (53e)$$

$$\alpha_c \lambda_a < \hat{\Delta}_b^{\hat{y}} + M_b - M_b \zeta_b^{\hat{y}-}, \forall \hat{y} \in \hat{\Omega}_y, \quad (53f)$$

$$\begin{aligned} \Phi_b(\mathbf{x}, \mathbf{y}) \geq & \Phi_b(\mathbf{x}, \hat{\mathbf{y}}) - M_a(\zeta_a^{\hat{y}+} + \zeta_a^{\hat{y}-}) \\ & - M_b(\zeta_b^{\hat{y}+} + \zeta_b^{\hat{y}-}), \forall \hat{y} \in \hat{\Omega}_y. \end{aligned} \quad (53g)$$

The main difference between **SLP** and **RSLP** is that we have replaced the set Ω_y in (52c)-(52g) with a new set $\hat{\Omega}_y$ in (53c)-(53g). Here $\hat{\Omega}_y \subseteq \Omega_y$ consists of samples of elements in Ω_y . Then we propose our constraint generation algorithm.

Note that problem **RSLP** (53) in step 2 is a relaxed version of problem **SLP** (52). Since problem (52), BMINLP (46) and the original trilevel optimization problem (21) are equivalent based on the previous analyses, the solution of **RSLP** leads to an upper bound of the optimal objective value. In step 3, the obtained $(\mathbf{x}_k, \mathbf{y}_k^m)$ is a feasible solution of BMINLP (46), then it is also a feasible solution of (21). Therefore, $\Phi_a(\mathbf{x}_k, \mathbf{y}_k^m)$ can serve as a lower bound of the optimal objective value. In step 4, $\mathbf{y}_k^m$ from step 3 is added to the sample set $\hat{\Omega}_y$ iteratively, which amounts to adding constraints to **RSLP**. Intuitively, performing step 4 in each iteration makes **RSLP** a better approximation to **SLP**, resulting in a tighter upper bound. Step 5 and 6 are the stopping criteria of this algorithm. If the condition of step 5 is not satisfied, step 7 and 8 update the lower bound and continue to the next iteration.

Remark 3: The proposed constraint generation algorithm is extended from [46, Algorithm 1] to handle equality constraints such as power balance equations (15).

Finally, we present a theorem about the exact convergence property of Algorithm 1.

Theorem 2: Under Assumption 1, the constraint generation algorithm converges finitely to a global optimal solution of the trilevel competitive pricing problem.

Proof: Please refer to Appendix B for the proof. ■

D. Linearization of RSLP & MLL

Despite that the trilevel competitive pricing problem has been reformulated into an iterative computation of single level optimization problems, problem **RSLP** at step 2 of Algorithm 1 and **MLL** at step 3 are still non-convex mixed-integer programs³ that are difficult to solve. For example, the upper level function $\Phi_a = (\lambda_a - \mu_a)(D_a - \alpha_s \lambda_a + \alpha_c \lambda_b)$ introduces three bilinear terms, which are $\alpha_s \mu_a \lambda_a$, $-\alpha_c \mu_a \lambda_b$ and $\alpha_c \lambda_a \lambda_b$, making Φ_a non-convex. To facilitate the computation, we seek to linearize the bilinear terms during implementation.

To be specific, based on Assumption 1, we rewrite the integer price variables $\bar{\lambda}_a$ and $\bar{\lambda}_b$ in binary variables. First, two vectors $r_a \in \mathbb{Z}_2^{\lfloor \log_2 K_a \rfloor + 1}$ and $r_b \in \mathbb{Z}_2^{\lfloor \log_2 K_b \rfloor + 1}$ are introduced. Here K_a and K_b denote the number of possible values of $\bar{\lambda}_a$ and $\bar{\lambda}_b$. The binary representation of $\bar{\lambda}_a$ and $\bar{\lambda}_b$ are as follows:

$$\bar{\lambda}_a = \sum_{i=0}^{\lfloor \log_2 K_a \rfloor} 2^i r_{ai}, \quad (54)$$

$$\bar{\lambda}_b = \sum_{i=0}^{\lfloor \log_2 K_b \rfloor} 2^i r_{bi}, \quad (55)$$

where r_{ai} and r_{bi} are the i -th elements of vector r_a and r_b , both of them are binary.

Next, we substitute the binary representation into bilinear terms. For instance, bilinear term $\lambda_a \mu_a$ becomes

$$\lambda_a \mu_a = \bar{\lambda}_a \mu_a s = s \sum_{i=0}^{\lfloor \log_2 K_a \rfloor} 2^i r_{ai} \mu_a.$$

For each r_{ai} , a new variable $w_{ai} = r_{ai} \mu_a$ is introduced. Then $\lambda_a \mu_a$ is equivalent to the following formulation:

$$\lambda_a \mu_a = s \sum_{i=0}^{\lfloor \log_2 K_a \rfloor} 2^i w_{ai},$$

where w_{ai} is constrained by

$$\begin{aligned} 0 & \leq \mu_a - w_{ai} \leq M_l(1 - r_{ai}), \\ 0 & \leq w_{ai} \leq M_l r_{ai}, \end{aligned}$$

and here M_l is a sufficiently large constant. Hence, the bilinear term $\lambda_a \mu_a$ has been transformed into linear formulations with respect to w_{ai} and r_{ai} . Other bilinear terms are linearized in similar ways. As a result, the non-convex mixed-integer programs **RSLP** and **MLL** have been transformed equivalently into MILPs.

³Even when the integer variables have been relaxed to continuous ones, the resulting problem is still non-convex.

E. Discussions

Our proposed solution methodology can be further extended to more general cases. For example, consider two EV charging networks (EVCN) A and B where each owns a set of EVCSs, i.e., $\mathcal{A} = \{1, 2, \dots, N_A\}$ and $\mathcal{B} = \{1, 2, \dots, N_B\}$. Denote the pricing decisions of EVCN A and B as $\vec{\lambda}_a = (\lambda_{a1}, \dots, \lambda_{aM})$ and $\vec{\lambda}_b = (\lambda_{b1}, \dots, \lambda_{bN})$. Suppose the demand functions in each station of EVCN A and B are $L_{am}(\vec{\lambda}_a, \vec{\lambda}_b)$, $m \in \mathcal{A}$ and $L_{bn}(\vec{\lambda}_a, \vec{\lambda}_b)$, $n \in \mathcal{B}$ respectively, where L_{am} and L_{bn} are non-linear functions. Then, if we replace the constants $\hat{\Delta}_a^{\hat{y}}$ and $\hat{\Delta}_b^{\hat{y}}$ defined in (50) and (51) with the following ones:

$$\begin{aligned}\hat{\Delta}_{am}^{\hat{y}} &= \hat{p}_{am} - d_{am}^p - \sum_{k:am \rightarrow k} \hat{P}_{am,k} - \sum_{i:i \rightarrow am} \hat{P}_{am,i}, \forall m \in \mathcal{A}, \\ \hat{\Delta}_{bn}^{\hat{y}} &= \hat{p}_{bn} - d_{bn}^p - \sum_{k:bn \rightarrow k} \hat{P}_{bn,k} - \sum_{i:i \rightarrow bn} \hat{P}_{bn,i}, \forall n \in \mathcal{B},\end{aligned}$$

after that we replace the constraints (52c)-(52f) in Problem SLP with the following inequalities:

$$\begin{aligned}L_{am}(\vec{\lambda}_a, \hat{\lambda}_b) &> \hat{\Delta}_{am}^{\hat{y}} - M_a + M_a \zeta_{am}^{\hat{y}+}, \forall \hat{y} \in \Omega_y, \forall m \in \mathcal{A}, \\ L_{am}(\vec{\lambda}_a, \hat{\lambda}_b) &< \hat{\Delta}_{am}^{\hat{y}} + M_a - M_a \zeta_{am}^{\hat{y}-}, \forall \hat{y} \in \Omega_y, \forall m \in \mathcal{A}, \\ L_{bn}(\vec{\lambda}_a, \hat{\lambda}_b) &> \hat{\Delta}_{bn}^{\hat{y}} - M_b + M_b \zeta_{bn}^{\hat{y}+}, \forall \hat{y} \in \Omega_y, \forall n \in \mathcal{B}, \\ L_{bn}(\vec{\lambda}_a, \hat{\lambda}_b) &< \hat{\Delta}_{bn}^{\hat{y}} + M_b - M_b \zeta_{bn}^{\hat{y}-}, \forall \hat{y} \in \Omega_y, \forall n \in \mathcal{B}.\end{aligned}$$

we can conclude that Theorem 1 and Theorem 2 still hold in this general case. The proofs are similar, thus are omitted.

IV. NUMERICAL RESULTS

In this section, we present numerical results about our proposed competitive pricing formulation and exact algorithm, and demonstrate their merits. The simulation is first performed on a small-scale distribution network using a 15-bus test case from [47], followed by a larger-scale distribution network using modified IEEE 123-bus test case. Finally, numerical results using a real-world test case taken from [48] are presented. All the simulations in this section are performed on a desktop with a 3.4 GHz Intel Core i5 CPU and 8 GB Memory. Problem **RSLP** in step 2 of Algorithm 1 and problem **MLL** in step 3 are solved with CPLEX 12.10 and MATLAB R2018b after transforming them into MILPs in Section III-D.

A. 15-Bus Case

The location of EVCS A is set at bus 4, while EVCS B is at bus 8. Suppose there are six zones with EV charging demands, where the intrinsic power demand parameters D are all set as 200 kW. The distances from six zones to EVCS A are $d_a^g = [2, 2, 2, 2, 5, 5]$ km, and distances to EVCS B are $d_b^g = [5, 5, 5, 5, 2, 2]$ km. The load-distance parameter w_d is assumed to be 0.2. The charging price cap is 2 CNY/kWh, and the price unit $s = 0.001$ CNY/kWh. Suppose there are two distributed generators at bus 5 and 12, the generation cost coefficients are set as $\mathbf{a}_g = [1, 0.5] \times 10^{-3}$ CNY/(kWh)², and $\mathbf{b}_g = [0.5, 0.4]$ CNY/kWh. Voltage magnitude upper and lower limits are assumed to be 1.05 p.u. and 0.95 p.u., respectively.

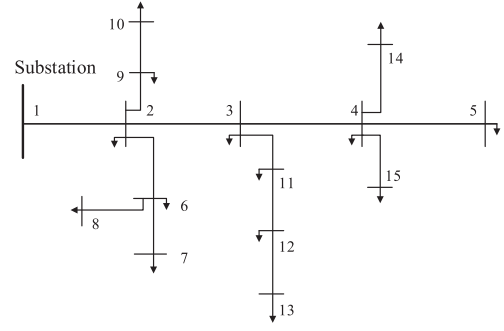


Fig. 3. 15-bus test case from [47].

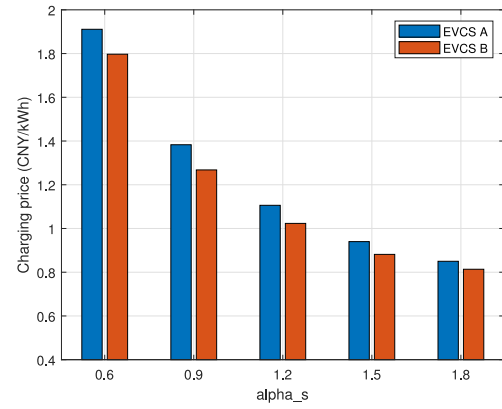


Fig. 4. Charging prices of EVCSs when α_s increases in the 15-bus case.

First, we vary the price sensitivity parameters α_s and α_c , where the self price sensitivity parameter α_s is increased from 0.6 to 1.8, the cross price sensitivity parameter is set as $\alpha_c = \alpha_s/2$. As shown in Fig. 4, when α_s increases, the charging prices of both EVCS A and EVCS B decrease. The reason is that when α_s becomes larger, the EV charging load becomes more elastic. Therefore, both EVCS A and B need to lower their prices to attract more price-sensitive EV users.

Next, we compare the profits of EVCS A using four different strategies, where the last three serve as benchmarks. These four strategies are elaborated as follows.

- *Optimal*: using our proposed trilevel model.
- *No-Comp*: EVCS A does not consider the potential competition from EVCS B. That is, A regards the charging price of B as a fixed value that does not change with A's price. In mathematical terms, EVCS A solves the following bilevel optimal pricing problem which only consists of upper level and lower level in (21):

$$\begin{aligned}\text{UL} : & \max_{\lambda_a} \Phi_a(\lambda_a, \lambda_b^*), \\ & \text{s.t. } 0 \leq \lambda_a \leq \lambda_{\max}, \\ \text{LL} : & \min_{\mathbf{u}} \sum_{k \in \mathcal{N}_g} C_k, \\ & \text{s.t. } (7)-(8), (13)-(19).\end{aligned}$$

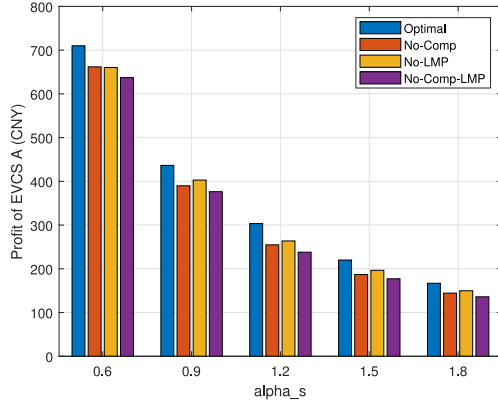


Fig. 5. EVCS A profit comparison using four different strategies in the 15-bus test case.

Here, λ_b^* is set as the optimal solution of EVCS B's own profit maximization problem as follows:

$$\begin{aligned}
 & \max_{\lambda_b} (\lambda_b - \mu_b)L_b, \\
 & \text{s.t. } 0 \leq \lambda_b \leq \lambda_{\max}, \\
 & \min_{\mathbf{u}} \sum_{k \in \mathcal{N}_g} C_k, \\
 & \text{s.t. (7)–(8), (13)–(19).}
 \end{aligned}$$

- *No-LMP*: EVCS A does not consider the impact of charging loads on the DLMP. In this case, EVCS A only solves its optimal pricing problem considering the competition from EVCS B based on the Stackelberg leader-follower game framework:

$$\begin{aligned}
 \text{UL} : & \max_{\lambda_a} \Phi_a(\lambda_a, \lambda_b), \\
 & \text{s.t. } 0 \leq \lambda_a \leq \lambda_{\max}, \\
 \text{ML} : & \max_{\lambda_b} \Phi_b(\lambda_a, \lambda_b), \\
 & \text{s.t. } 0 \leq \lambda_b \leq \lambda_{\max},
 \end{aligned}$$

where DLMP μ_a and μ_b are set as constant parameters. These constant DLMPs are computed by solving the OPF problem in advance, assuming the charging loads at EVCS A and B are divided proportionally according to their distances to the charging demands.

- *No-Comp-LMP*: EVCS A considers neither the price competition from EVCS B, nor the impact on the DLMP, i.e., solving the following problem:

$$\begin{aligned}
 \text{UL} : & \max_{\lambda_a} \Phi_a, \\
 & \text{s.t. } 0 \leq \lambda_a \leq \lambda_{\max},
 \end{aligned}$$

where fixed parameters λ_b and μ_a are set in the similar way with the No-Comp and No-LMP cases.

In Fig. 5, compared with other three benchmark strategies, our proposed trilevel model achieves the largest profit for EVCS A. This proves the necessity of considering the effect of competition between EVCSs, as well as the importance of taking impacts on the DLMP into account. Hence, our trilevel competitive pricing model has been validated.

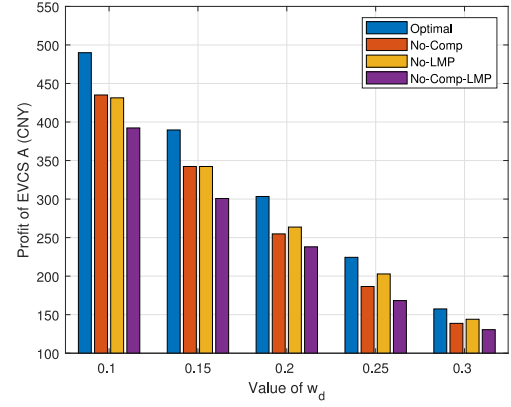


Fig. 6. EVCS A profit when w_d increases.

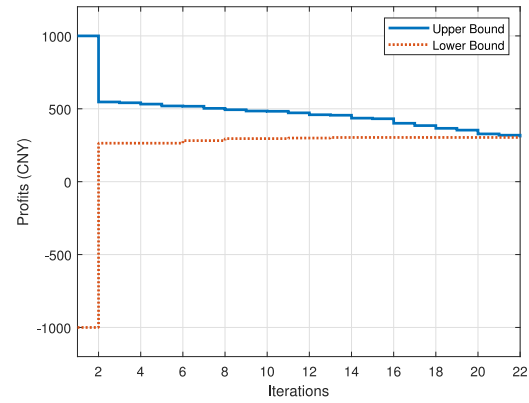


Fig. 7. Convergence of Algorithm 1 in the 15-bus case.

After that, we vary the value of load-distance parameter w_d . The self-price sensitivity parameter α_s is set as 1.2. As shown in Fig. 6, under different values of w_d , our proposed framework still brings larger profit for EVCS A than other three benchmarks. Therefore, the generality of the proposed framework with respect to w_d has been demonstrated.

Finally, we illustrate the convergence property of Algorithm 1 by simulation. The case with $\alpha_s = 1.2$ is taken as an example, while other cases can achieve similar results. In Fig. 7, the upper bound is updated every step, as in step 2 of Algorithm 1. At the same time, the lower bound is either updated or kept unchanged, as in step 8 or 10 of Algorithm 1. When the upper and lower bound converge after iteration 21, where the average computation time of each iteration is 0.93s, the algorithm terminates. This conforms with Theorem 2 that Algorithm 1 converges in finite iterations.

B. IEEE 123-Bus Case

In this subsection, we extend our simulations to a larger network. Suppose the IEEE 123-bus distribution network in Fig. 8 is coupled with the Sioux-Falls transportation network in Fig. 9. Specifically, each of the green centroids in Sioux-Falls network is at a certain bus of the IEEE 123-bus network.

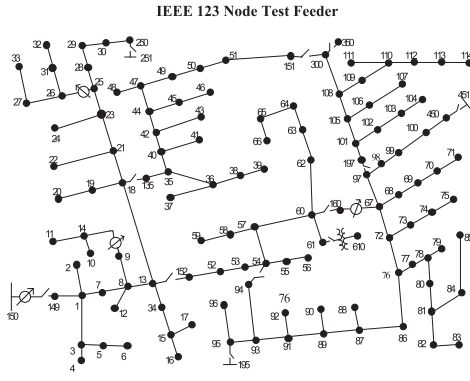


Fig. 8. IEEE 123-bus test case.

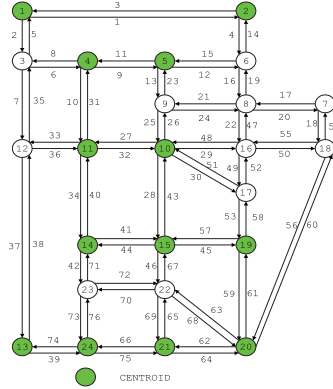


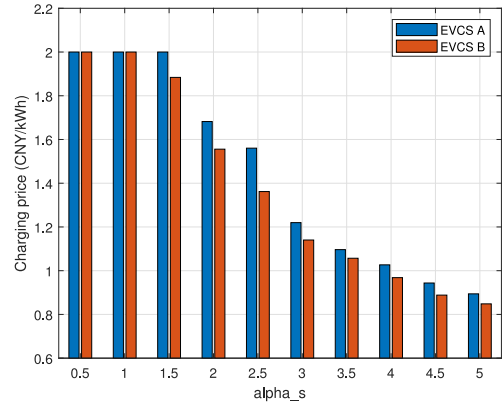
Fig. 9. Sioux-Falls transportation network.

TABLE II
COUPLING RELATIONSHIP BETWEEN SIOUX-FALLS
NETWORK AND IEEE 123 BUS NETWORK

S-F	123B	S-F	123B	S-F	123B
1	110	11	66	20	13
2	76	13	25	21	1
4	101	14	47	24	18
5	67	15	35		
10	60	19	54		

Detailed coupling relationships are shown in Table II. EVCS A is assumed to be at node 19, according to Table II, it is at bus 54. EVCS B is assumed to be at node 1, i.e., bus 110. The distance between two nodes in the transportation network is set as the length of the shortest path between them, while the arc length between two neighboring nodes is assumed to be 1. Suppose there are seven distributed generators located at bus 20, 36, 47, 52, 69, 91 and 112, respectively. Generation cost coefficients are assumed to be $\mathbf{a}_g = [0.05, 0.1, 0.03, 0.05, 0.05, 0.03, 0.1] \times 10^{-3}$ CNY/(kWh)², and $\mathbf{b}_g = [0.1, 0.3, 0.15, 0.1, 0.2, 0.05, 0.4]$ CNY/kWh. Other parameters for this simulation follow from [49].

We first vary the price sensitivity parameters α_s and α_c . Here α_s is increased from 0.5 to 5. α_c is set as the half of α_s . In Fig. 10, the charging price decreases as EV charging loads becomes more elastic, which is similar with the result in 14-bus case. Note that the charging price of EVCS A is in general higher than that of EVCS B except for the first two cases. The reason is that the average distance from zones to A

Fig. 10. Charging prices of EVCS A and B when α_s increases in the 123-bus case.

is less than that of B. In fact, the average distance from zones to A is 2.875, while the distance to B is 3.625. Therefore, EVCS A is generally more attractive to the charging demands than B, resulting in a higher charging price. When $\alpha_s = 0.5$ and 1, charging prices are all equal to the price cap. This is because when charging loads are rather inelastic, i.e., EV users have no other charging alternatives, they have to accept the charging price up to the price cap.

After that, we compare the profits of EVCS A under four different strategies discussed before: *Optimal*, *No-Comp*, *No-LMP*, *No-Comp-LMP*. As shown in Fig. 11(a) and Fig. 11(b), considering both the price competition from EVCS B and the impact of charging loads on DLMP achieves the largest profit for EVCS A. This result further validates our proposed model.

In Fig. 12, the convergence result of Algorithm 1 is demonstrated. The price sensitivity parameter α_s is set as 3 here. Other values of α_s achieve similar results. We can see from Fig. 12 that Algorithm 1 converges after 34 iterations. The corresponding computation time of each iteration is also depicted in Fig. 13. With Algorithm 1 iterates, more samples are added to the sample set $\hat{\Omega}_y$, which means more constraints are added to problem **RSLP**, thus the computation time of each iteration has an increasing trend. Next, the computational properties of Algorithm 1 when parameter α_s varies are presented in Table III. The second column is the number of iterations required for convergence, the third column is the average computation time of each iteration, while the last column is total computation time. We can conclude that for the larger test case, the computational load of our proposed framework is still acceptable.

Finally, the location of EVCS A is changed to show the potential profit it can obtain in different places. The result is summarized in the following Table IV. The unit of profit is in Chinese Yuan (CNY). From this table, we can conclude that at location 10, EVCS A can obtain the largest profit. Therefore, our competitive pricing model can serve as a guidance in the location selection process of new EVCS.

C. Real-World Map Based Test Case

We perform another set of simulations using the real-world test case from [48], which is adapted from the real map of

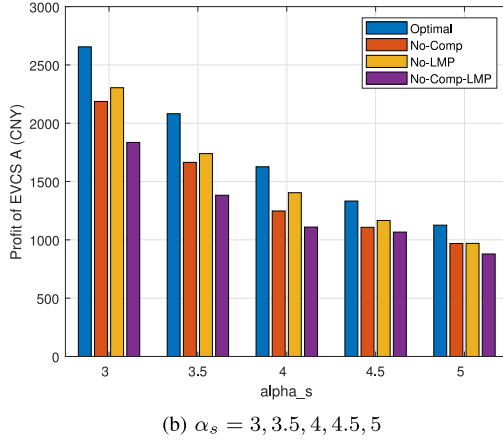
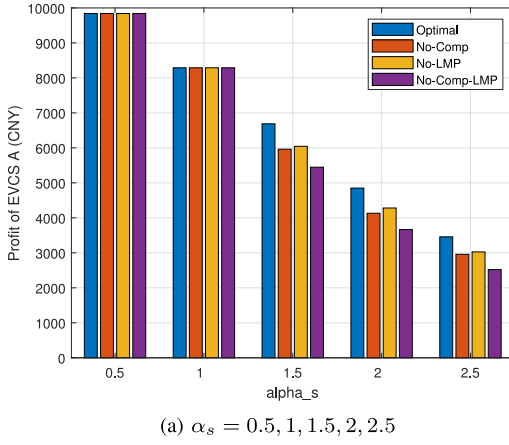


Fig. 11. EVCS A profit comparison using four different strategies in the 123-bus test case.

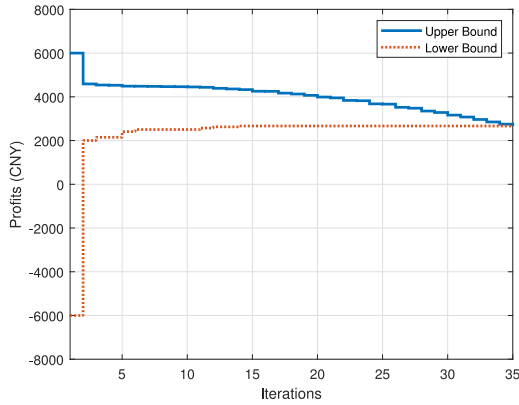


Fig. 12. Convergence of Algorithm 1 in the 123-bus case.

Luohu district, Shenzhen city, China. According to [48], this test case consists of a 47-intersection transportation system and 33-node distribution system. These two systems are presented in Fig. 14 and Fig. 15, respectively. Similar with the setup in [48], we assume the locations of EVCS A and B are at road intersections 9 and 40, which are connected to bus 23 and 26, respectively. Besides that, charging demands are supposed to be from intersection 1, 2, 19, 27, 28, 33, 36 and 42. The

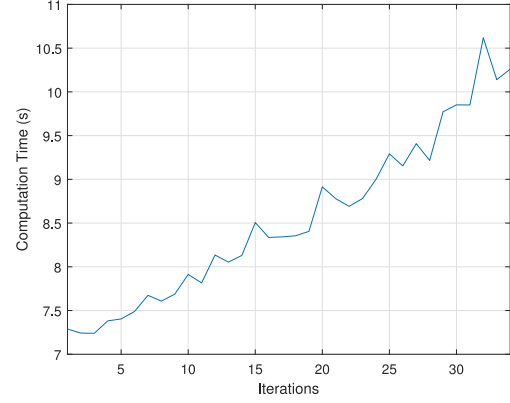


Fig. 13. Computation time of each iteration using Algorithm 1.

TABLE III
COMPUTATIONAL PROPERTIES OF ALGORITHM 1 WHEN α_s VARIES

α_s	iterations	mean time (s)	total time (s)
0.5	1	6.53	6.53
1	1	7.12	7.12
1.5	4	7.42	29.67
2	25	8.06	201.44
2.5	33	8.34	275.24
3	34	8.55	290.73
3.5	33	8.22	271.17
4	33	8.33	274.97
4.5	31	8.26	256.13
5	30	8.14	244.15

TABLE IV
PROFITS OF EVCS A AT DIFFERENT LOCATIONS

Node	Profit	Node	Profit	Node	Profit
2	1816.36	11	2590.15	19	2655.18
4	2468.12	13	2492.86	20	2517.04
5	2312.44	14	2361.22	21	2535.28
10	2766.74	15	2664.35	24	2455.65

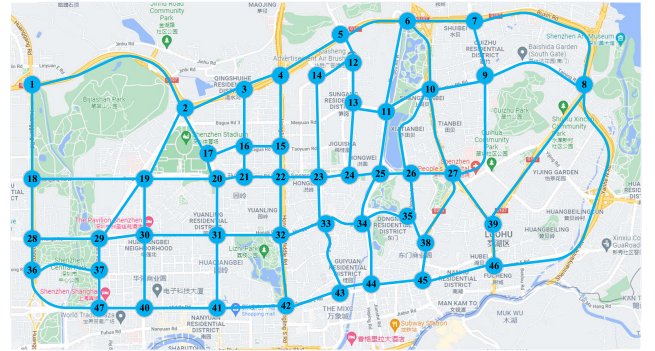


Fig. 14. The 47-intersection transportation system adapted from Luohu district, Shenzhen city, China.

distance between two intersections is calculated as the shortest geographic distance along the road.

Like the last two set of simulations, we first present the EVCS charging prices with respect to α_s , which is in Fig. 16. It can be concluded that charging prices also decrease when charging demands become more elastic.

After that, the profits of EVCS A achieved by our proposed framework are compared with other three benchmarks: *No-Comp*, *No-LMP* and *No-Comp-LMP*. As shown in

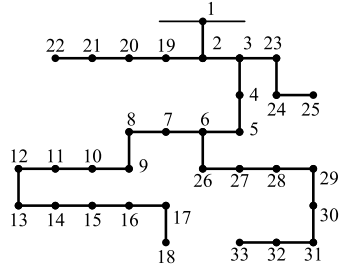


Fig. 15. 33-node distribution system [48].

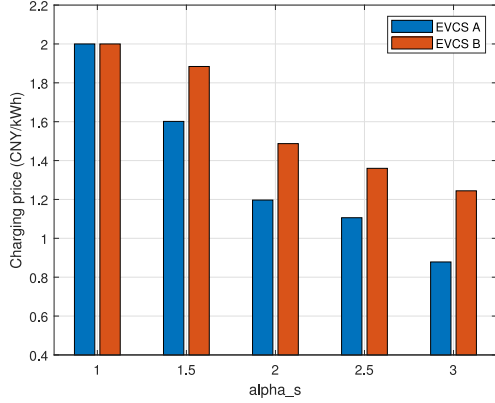
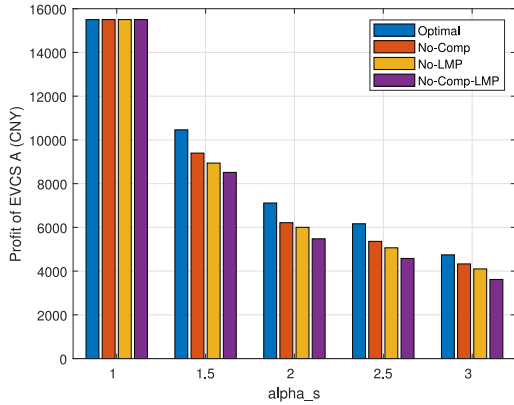
Fig. 16. Charging prices of EVCS A and B with respect to α_s .

Fig. 17. Profits of EVCS A under four different strategies.

Fig. 17, among four different strategies, our proposed one is the most profitable one for EVCS A.

Finally, the convergence property of Algorithm 1 in this test case is illustrated by Fig. 18. α_s in this scenario is set as 2. In Fig. 18, the algorithm converges after iteration 28, while the average computation time of each iteration is 2.76s. This demonstrates that our proposed algorithm converges in finite iterations, and the computation time is also acceptable.

V. CONCLUSION & FUTURE WORKS

In this work, we studied the competitive pricing problem of EVCS in a coupling transportation-power network. The optimal pricing strategy of EVCS was determined, taking into account both the competition from other EVCS and the

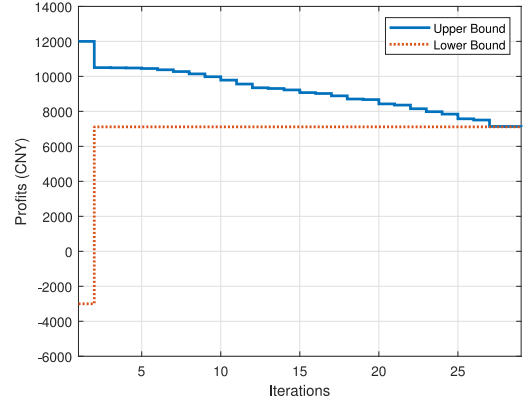


Fig. 18. Convergence of Algorithm 1.

impact of charging loads on the DLMP. A trilevel optimization model was proposed, where the upper level was the profit maximization problem of this EVCS, middle level captured its competitor's profit maximization problem, while the lower level represented the OPF problem. As the trilevel model is challenging to solve, it was reformulated into a bilevel mixed-integer program first, and then reformulated into a single-level one. Equivalence of these problems was rigorously proved. Furthermore, we developed an exact algorithm to compute the global optimal solution in finite iterations. Finally, numerical results validated our trilevel model and exact algorithm, which demonstrate their advantage of increasing profits for EVCS, and shows their potential to guide future EVCS's siting.

In the future, we plan to extend this work in two directions. First, charging demand uncertainty due to random traffic conditions will be accounted for based on stochastic optimization methods such as sample average approximation. Second, we can extend the model to the multi-leader multi-follower price competition case, where the profit-maximizing problems in the upper two levels will be replaced by non-cooperative games.

APPENDIX A PROOF OF THEOREM 1

Since it is easy to see that for $(\mathbf{x}, \mathbf{y})$, satisfying (46b) and (46d) is equivalent to satisfying (52b), we focus on the equivalence between (46c) and (52c)-(52g) here.

Constraint (46c) requires lower level optimality for the bilevel program, as shown in (48). This constraint can also be represented by the following inequality:

$$\Phi_b(\mathbf{x}, \mathbf{y}) \geq \Phi_b(\mathbf{x}, \tilde{\mathbf{y}}), \forall \tilde{\mathbf{y}} \in \Omega_y(\mathbf{x}). \quad (56)$$

Examining the definition of $\hat{\Delta}_a^{\hat{\mathbf{y}}}$ and $\hat{\Delta}_b^{\hat{\mathbf{y}}}$ in (50) and (51) yields that if $\alpha_s \lambda_a = \hat{\Delta}_a^{\hat{\mathbf{y}}}$ and $\alpha_c \lambda_a = \hat{\Delta}_b^{\hat{\mathbf{y}}}$, then

$$\begin{aligned} \hat{p}_a - d_a^p - (D_a - \alpha_s \lambda_a + \alpha_c \hat{\lambda}_b) &= \sum_{k:a \rightarrow k} \hat{P}_{ak} + \sum_{i:i \rightarrow a} \hat{P}_{ai}, \\ \hat{p}_b - d_b^p - (D_b - \alpha_s \hat{\lambda}_b + \alpha_c \lambda_a) &= \sum_{k:b \rightarrow k} \hat{P}_{bk} + \sum_{i:i \rightarrow b} \hat{P}_{bi}, \end{aligned}$$

which mean that an $(\mathbf{x}, \hat{\mathbf{y}})$ is feasible for the lower level problem **MLL**, i.e., $(\mathbf{x}, \hat{\mathbf{y}}) \in \Omega$ or $\hat{\mathbf{y}} \in \Omega_y(\mathbf{x})$. Thus, (56)

means that an feasible solution of BMINLP (46) $(\mathbf{x}, \mathbf{y})$ must satisfy $\Phi_b(\mathbf{x}, \mathbf{y}) \geq \Phi_b(\mathbf{x}, \hat{\mathbf{y}})$.

First we consider a feasible solution $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ of the single-level optimization problem **SLP** (52). Note that $\mathbf{z} = (\zeta_a^{\hat{\mathbf{y}}+}, \zeta_a^{\hat{\mathbf{y}}-}, \zeta_b^{\hat{\mathbf{y}}+}, \zeta_b^{\hat{\mathbf{y}}-})$ for all $\hat{\mathbf{y}} \in \Omega_y$, where values of $\zeta_a^{\hat{\mathbf{y}}+}$ and so on must be zero or one. We discuss four possible value combinations for $\zeta_a^{\hat{\mathbf{y}}+}$ and $\zeta_a^{\hat{\mathbf{y}}-}$:

- $\zeta_a^{\hat{\mathbf{y}}+} = 1$ and $\zeta_a^{\hat{\mathbf{y}}-} = 0$: Here, (52g) becomes redundant. By (52c) and (52d), we have $\alpha_s \lambda_a > \hat{\Delta}_a^{\hat{\mathbf{y}}}$, according to the discussion above, $\hat{\mathbf{y}} \notin \Omega_y(\mathbf{x})$, in this case, lower level optimality constraint (56) also does not exist in the BMINLP.
- $\zeta_a^{\hat{\mathbf{y}}+} = 0$ and $\zeta_a^{\hat{\mathbf{y}}-} = 1$: This case corresponds to $\alpha_s \lambda_a < \hat{\Delta}_a^{\hat{\mathbf{y}}}$, similar to the first case.
- $\zeta_a^{\hat{\mathbf{y}}+} = 1$ and $\zeta_a^{\hat{\mathbf{y}}-} = 1$: This kind of values lead to the contradiction between (52c) and (52d), thus is not a feasible solution for problem (52).
- $\zeta_a^{\hat{\mathbf{y}}+} = 0$ and $\zeta_a^{\hat{\mathbf{y}}-} = 0$: (52g) becomes $\Phi_b(\mathbf{x}, \mathbf{y}) \geq \Phi_b(\mathbf{x}, \hat{\mathbf{y}}), \forall \hat{\mathbf{y}} \in \Omega_y$. By (49), $\Omega_y(\mathbf{x}) \subseteq \Omega_y$, thus the feasible region of (52) is no larger than that of BMINLP.

The cases with $\zeta_b^{\hat{\mathbf{y}}+}$ and $\zeta_b^{\hat{\mathbf{y}}-}$ are similar. Therefore, to conclude, for a feasible solution $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ of **SLP** (52), the corresponding $(\mathbf{x}, \mathbf{y})$ must also be a feasible solution of the BMINLP.

Next we consider a feasible solution of BMINLP $(\mathbf{x}, \mathbf{y})$ and seek to find $\mathbf{z}$ that makes $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ a feasible solution to (52). Three cases are discussed as follows:

- $\alpha_s \lambda_a < \hat{\Delta}_a^{\hat{\mathbf{y}}}$: Lower level optimality constraint (56) does not exist in the BMINLP by definition. According to (52c) and (52d), $\zeta_a^{\hat{\mathbf{y}}+} = 0$ while $\zeta_a^{\hat{\mathbf{y}}-}$ is unconstrained. Let $\zeta_a^{\hat{\mathbf{y}}-} = 1$, then constraint (52g) also becomes redundant in the single level reformulation.
- $\alpha_s \lambda_a > \hat{\Delta}_a^{\hat{\mathbf{y}}}$: This case is similar to the first one.
- $\alpha_s \lambda_a = \hat{\Delta}_a^{\hat{\mathbf{y}}}$: In this case, $(\mathbf{x}, \hat{\mathbf{y}})$ is feasible, i.e., $\hat{\mathbf{y}} \in \Omega_y(\mathbf{x})$. By (52c) and (52d), both $\zeta_a^{\hat{\mathbf{y}}+}$ and $\zeta_a^{\hat{\mathbf{y}}-}$ must be zero, making (52g) an active constraint for all $\hat{\mathbf{y}} \in \Omega_y(\mathbf{x})$, which is equivalent to the lower level optimality constraint (56) of BMINLP.

The cases with $\alpha_c \lambda_a$ and $\hat{\Delta}_b^{\hat{\mathbf{y}}}$ are also similar. Therefore, for every feasible solution of BMINLP, we can find a corresponding feasible solution for the single-level reformulation.

Finally, notice that the objective function of BMINLP and (52) are exactly the same, so the theorem is proved.

APPENDIX B PROOF OF THEOREM 2

Under Assumption 1, Ω_x is a finite set. Suppose that Algorithm 1 has completed $|\Omega_x| + 1$ iterations, then there must exist two iterations i and j , $1 \leq i < j \leq |\Omega_x| + 1$, such that $\mathbf{x}_i = \mathbf{x}_j$. In step 4 of Algorithm 1, a $\mathbf{y}_i^m$ which is the optimal solution of lower level problem **MLL**, i.e., $\mathbf{y}_i^m \in \arg\max\{\Phi_b | \mathbf{y} \in \Omega_y(\mathbf{x}_i)\}$, is added to the sample set $\hat{\Omega}_{y,i+1}$. Due to $\hat{\Omega}_{y,i+1} \subseteq \hat{\Omega}_{y,j}$, then we have $\mathbf{y}_i^m \in \hat{\Omega}_{y,j}$. Furthermore, $\mathbf{y}_i^m \in \Omega_y(\mathbf{x}_i)$ and $\Omega_y(\mathbf{x}_i) = \Omega_y(\mathbf{x}_j)$ lead to $\mathbf{y}_i^m \in \Omega_y(\mathbf{x}_j)$, which means $(\mathbf{x}_j, \mathbf{y}_i^m)$ is feasible for the BMINLP. In this case, the lower optimality constraint (53g) is active.

Combining $\mathbf{y}_i^m \in \hat{\Omega}_{y,j}$, $\mathbf{y}_i^m \in \Omega_y(\mathbf{x}_j)$ and inequality (53g), we have $\Phi_b(\mathbf{x}_j, \mathbf{y}_i^m) \geq \Phi_b(\mathbf{x}_j, \mathbf{y}_j^m)$. As $\mathbf{y}_i^m \in \arg\max\{\Phi_b | \mathbf{y} \in \Omega_y(\mathbf{x}_j)\}$ due to $\Omega_y(\mathbf{x}_i) = \Omega_y(\mathbf{x}_j)$, this implies $\mathbf{y}_j^m$ also satisfies the lower level optimality condition, i.e., $\mathbf{y}_j^m \in \arg\max\{\Phi_b | \mathbf{y} \in \Omega_y(\mathbf{x}_j)\}$. Therefore, we have $\Phi_b(\mathbf{x}_j, \mathbf{y}_j^m) = \Phi_b(\mathbf{x}_j, \mathbf{y}_i^m)$. Algorithm 1 goes to step 5 and 6, and converges at iteration j with solution $(\mathbf{x}_j, \mathbf{y}_j^m)$. Since this solution for relaxed problem is now feasible for the original problem, as well as $j \leq |\Omega_x| + 1$, we can claim that Algorithm 1 converges within finite steps to a global optimal solution.

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